

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester-IV Examination, 2022-23

Botany
Paper: BOB-401 — Phanerogams

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two questions each from Section A and B. Question No. 1 is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Briefly answer the following in not more than 50 words each: [1 × 10 = 10]

- (a) Define pycnoxylic wood.
- (b) Differentiate between centripetal and centrifugal xylem.
- (c) What is sulphur shower?
- (d) Define megasporogenesis.
- (e) Write the systematic position of *Ephedra*.
- (f) Write the primitive characters of family Ranunculaceae.
- (g) Opium drug is obtained from which plant?
- (h) Define Pollinia.
- (i) What is stylopodium?
- (j) Write the name of a plant that possesses adnate stipule.

SECTION A

Question 2. Describe in detail the reproduction in *Cycas* along with suitable diagrams. [15]

Question 3. Discuss any two of the following along with well-labelled diagrams: [7.5 × 2 = 15]

- (a) Anatomy of *Cycas* leaflet
- (b) Anatomy of *Ephedra* stem
- (c) Anatomy of *Pinus* needle

Question 4. Give a comparative account of anatomical features of *Pinus* and *Gnetum* stem. [15]

Question 5. Differentiate any two of the following: [7.5 × 2 = 15]

- (a) Male cone of *Cycas* and *Pinus*

- (b) Microsporophyll and Megasporophyll of *Cycas*
- (c) Anemophily and Entomophily

SECTION B

Question 6. Discuss the floral characteristics, systematic position and economic importance of family Fabaceae. [15]

Question 7. Describe the various types of endosperm development in Angiosperms with suitable diagrams. [15]

Question 8. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Aggregate fruits
- (b) Verticillaster inflorescence
- (c) Double fertilization

Question 9. Define pollination. Describe the various contrivances for cross-pollination. [15]

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